Sri Lanka Institute of Information Technology

IT1040 - Fundamentals of Computing
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A Learning Tool for Children

Proposal Document

Group WD.01.40

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1. Background

Smart plant buddy is about helping children learn how to take care of plants while also teaching them basic science ideas, like sunlight, temperature, & water. We are creating a smart plant monitoring system that connects with a fun and interactive website. This website will show live updates from the plant using sensors and also give simple tips and animations to help them understand what the plant needs. The goal is to make learning about nature and science fun, simple and hands-on for young children.

2. Problem & Motivation

2.1 Problem

In today's digital age, many children have limited exposure to nature and practical science. They often use phones, computers, or tablets, but they don't care about them. Science is sometimes taught in a way that is boring or hard to understand, because these children may not learn important things like how sunlight, water, and temperature help plants live and grow.

2.2 Motivation

We want to help children learn about nature and science in a fun and easy way. Our idea is to make a smart system that shows how a plant is doing using real data from sensors. We will connect this to a colorful & friendly website where kids can learn about plants & science through simple words & pictures. This will help them enjoy learning and also teach them how to care for plants in real life.

3. Aim and Objectives

3.1 Aim

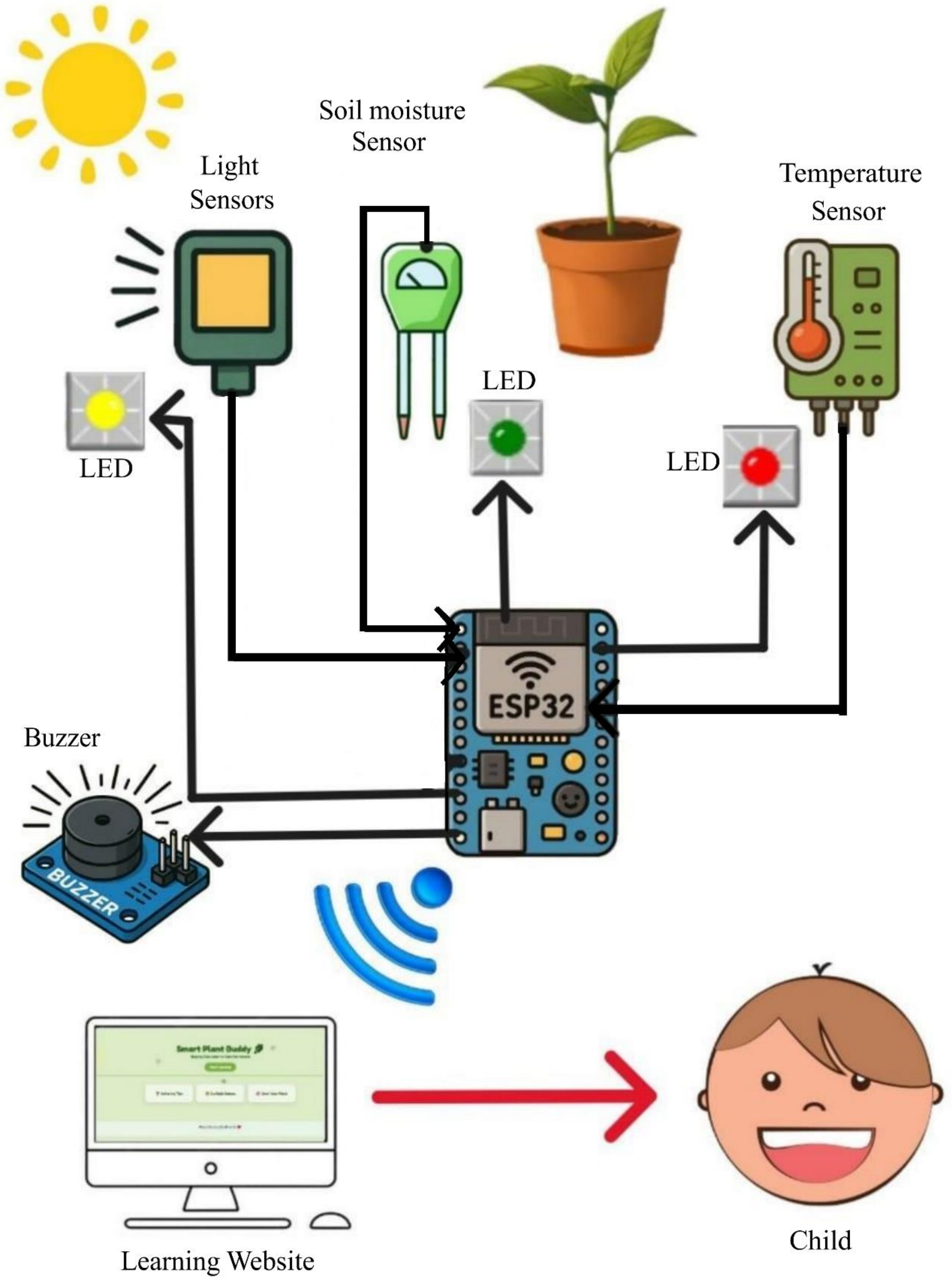
The aim of our project is to build a smart plant monitoring system with a fun & easy to use website that helps children learn how to care for plants & understand basic science topics like light, water & temperature.

3.2 Objectives

To achieve this aim, we plan to,

1. Use sensors to check the light, temperature & water levels
2. Send the sensor data to a website in real time.
3. Design a simple & colorful website for children to use.
4. Show helpful messages, tips, and pictures based on sensor data.
5. Help children learn basic science ideas through the website in an easy and interesting way.

4. System diagram



5. <u>Methodology</u>

1. Hardware Component

- **Microcontroller – ESP32**

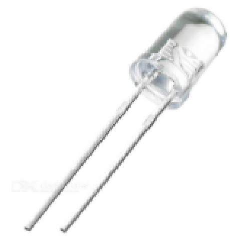
- **Sensors** - **Soil Moisture Sensor** (To detect when the plant needs water.)

- **Light sensor (LDR**) (To check if the plant gets enough sunlight)

- **Temperature Sensor** (To monitor the surrounding conditions.)

- **Actuators/ Indicators – LED**

  (When the plant needs water, sunlight, or a suitable temperature, it glows blue, yellow, and red LED bulbs, respectively. And produces a simple tone with the buzzer)



- **Buzzer**

(To produce an audible sound, acting as an indicator.)

- **Memory - Data Base**

2. Software Side

- **Web Interface**

6. Evaluation Methods

- Sensor accuracy testing
  - Checking the sensors separately.

- Website evaluation.

- Test if the system gives alerts when needed
  - Checking if the system gives the correct output for the plant's needs.

- Learning outcome assessment

  - Ask simple questions after using the system.

  - Measure if they learned something new.

- Check website design and daily display.

7. References

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